

Theories of Computation Formative Assignment 22/23

Question 1 Give a regular expression over the alphabet $\Sigma = \{a, b\}$ for the set of words in which the substring ' ab ' does not occur. For example, "baa" will be accepted but " aab " will not. [2 marks]

Question 2 Consider the regular expression $(a|b^*)^*a$. In a step-by-step fashion, using the rules from Handout 2b, convert each sub-expression of the regular expression into an ϵ NFA and, by doing so, obtain a DFA that is equivalent to the overall regular expression. [5 marks]

Question 3 Consider the following sequence of natural numbers $D = \{1, 2, 3, 6, 11, 20, \dots\}$ defined as $D(0) = 1$, $D(1) = 2$, $D(2) = 3$ and $D(n+3) = D(n) + D(n+1) + D(n+2)$. Show, by induction, that for any natural number n , $D(n) < 2^n$. [3 marks]